



MERIDIAN JUNIOR COLLEGE
JC 2 Preliminary Examination
 H2 Chemistry Paper 4

1 (a) Procedure

Burette readings

Final burette reading / cm ³	48.00
Initial burette reading / cm ³	0.00
Volume of FA 2 / cm ³	48.00

Mass Readings

Mass of weighing bottle and FA 1 / g	6.789
Mass of weighing bottle and residual solid / g	4.789
Mass of FA 1 added / g	2.000

Titration

	1	2
Final burette reading / cm ³	21.30	21.30
Initial burette reading / cm ³	0.00	0.00
Volume of FA 3 / cm ³	21.30	21.30

✓

✓

(b) Average volume of FA3 used = $\frac{21.30 + 21.30}{2} = 21.30 \text{ cm}^3$

(c) (i)

Amount of HCl in FA3 in 21.30 cm³
 $= \frac{25.0}{1000} \times 0.2 = 5.00 \times 10^{-3} \text{ mol}$

(ii) Amount of HCl remaining in FA3 (in 250 cm³)

$= \frac{250}{17.50} \times 5 \times 10^{-3} = 0.0587 \text{ mol}$

(iii) Initial amount of HCl = $\frac{48.00}{1000} \times 2.0 = 0.0960 \text{ mol}$

Amount of HCl in FA3 that has reacted = $0.0960 - 0.0587$
 $= 0.0373 \text{ mol}$

(iv) Amount of $\text{MgCO}_3 = \frac{0.0373}{2}$
 $= 0.01865 \text{ mol}$

Mass of $\text{MgCO}_3 = 0.01865 \times (24.3 + 12 + 48) = 1.57 \text{ g}$

Percentage purity of FA1 = $\frac{1.57}{2.00} \times 100\% = 78.6\%$

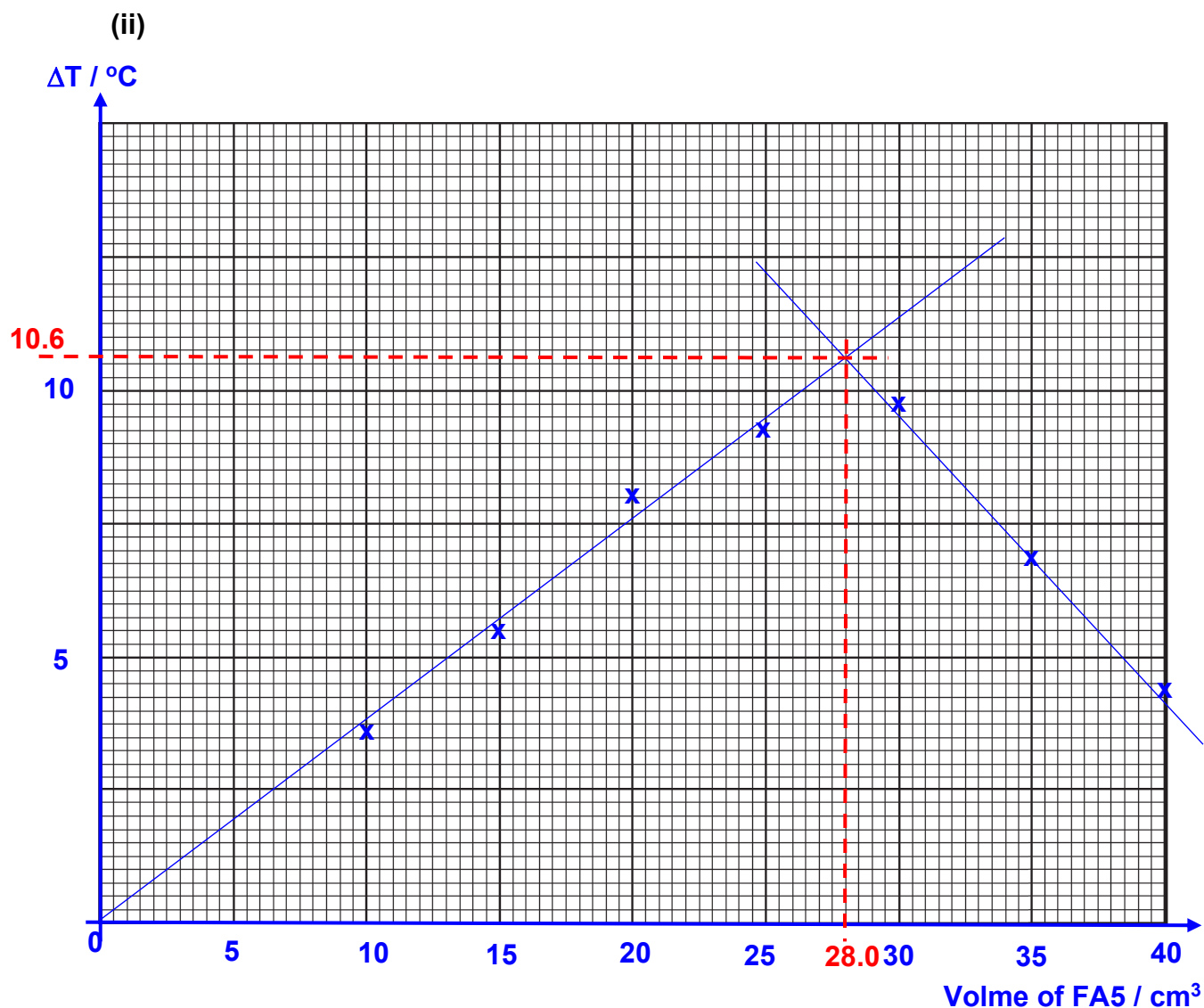
(d) (i) smallest volume used = 48.40 cm^3
 largest volume used = 48.60 cm^3

(ii) The percentage purity of FA 1 should be the same for student X and student Y since FA2 (hydrochloric acid) is added in excess to completely react with FA1 (magnesium carbonate).

2 Results

Expt	Volume of FA 5 / cm^3	Volume of FA 6 / cm^3	$T_{\text{FA5}} / ^\circ\text{C}$	$T_{\text{FA6}} / ^\circ\text{C}$	$T_{\text{weighted initial}} / ^\circ\text{C}$	$T_{\text{max}} / ^\circ\text{C}$	$\Delta T / ^\circ\text{C}$
1	10.0	40.0	28.0	32.0	31.2	34.5	3.3
2	20.0	30.0	30.0	32.0	31.2	38.0	6.8
3	30.0	20.0	31.0	31.5	31.2	41.0	9.8
4	40.0	10.0	31.0	31.5	31.1	35.5	4.4
5	15.0	35.0	29.5	31.5	30.9	36.0	5.1
6	25.0	25.0	30.5	31.5	31.0	40.0	9.0
7	35.0	15.0	31.0	31.5	31.2	38.0	6.8

[5]



(iv) $\Delta T_{\max} = 10.6 ^\circ\text{C}$ $V_{\max} = 28.0 \text{ cm}^3$

(b) (i) No of moles of KOH = $[(50 - 28) / 1000] \times 1.50 = 0.0330 \text{ mol}$

No of moles of $\text{H}_2\text{SO}_4 = 0.0375 \times \frac{1}{2} = 0.0165 \text{ mol}$

$$[\text{H}_2\text{SO}_4] \text{ in FA 5} = \frac{0.0165}{\frac{28.00}{1000}} = 0.589 \text{ mol dm}^{-3}$$

(ii) $Q = (50)(4.18)(10.6) = 2215 \text{ J}$

(c) $\Delta H_n = - \frac{2215}{2 \times [0.589] \times \frac{28.00}{1000}} = -67.1 \text{ kJ mol}^{-1}$

- (d) As the temperature rises to the maximum value, there is a greater temperature difference between the solution and the surrounding. Hence, there is greater heat loss to the surrounding or there is a greater temperature gradient.
Hence, the temperature is more inaccurate as they approach the maximum value.

- (e) **for graph**

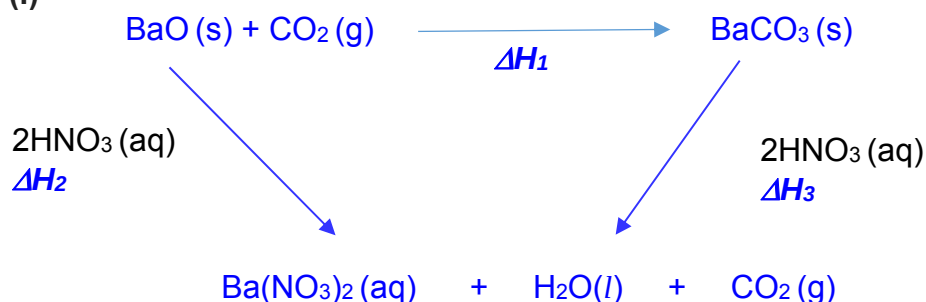
Maximum is reached at the same V_{max} of sulfuric acid as the experiment with potassium hydroxide.

for explanation

Lower temperature rise, ΔT_{max} , as some heat that would have been released is used to dissociate the weak base completely. Maximum is reached at the same number of moles of sulfuric acid as the reacting mole ratio remains the same. Hence, V_{max} is the same.

- (f) **Planning**

(i)



By Hess Law, $\Delta H_1 = \Delta H_2 - \Delta H_3$

(ii) Pre-calculations

If mass of BaO used is 1.00 g, amount of BaO = $\frac{1.00}{137.0+16.0} = 0.006536 \text{ mol}$

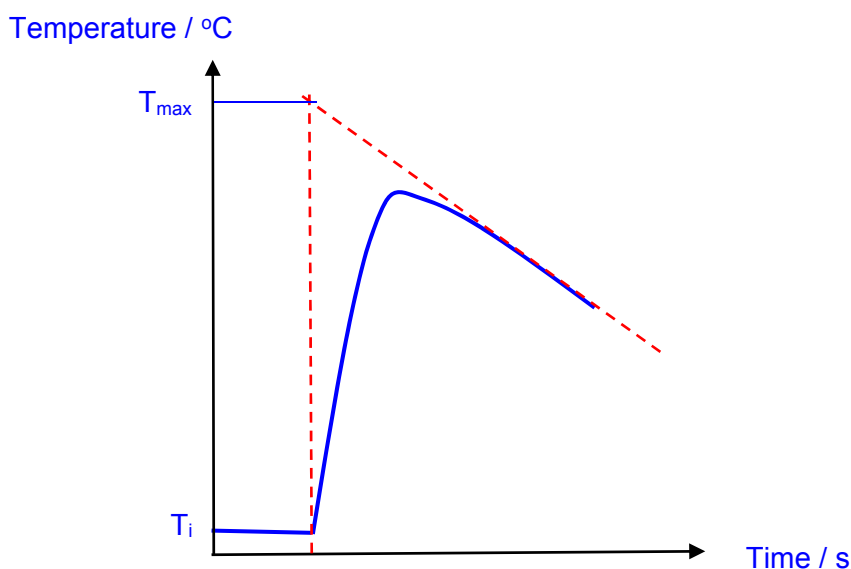
Amount of HNO₃ = 2 x 0.006536 = 0.01307 mol

Minimum volume of HNO₃ required to dissolve the BaO solid = $\frac{0.01307}{0.500} = 26.14 \text{ cm}^3$

Experimental Procedure

1. Weigh out about 1 g of BaO into a dry weighing bottle. Record the mass readings in a table.
2. Using a 50 cm³ measuring cylinder, transfer 40 cm³ of HNO₃ into the styrofoam cup supported in a glass beaker.
3. Measure and record the initial temperature of the HNO₃. Start the stop-watch and stir the solution gently with the thermometer. Record the temperature of the solution every 30 seconds for 2.0 minutes.
4. At 2.0 minutes, transfer the BaO solid into the styrofoam cup. Continuously stir the mixture thoroughly, and read the temperature at 2.5 minutes. Following this, record the temperature every 30 seconds from 3.0 minutes to 8.0 minutes.
5. At the end of the experiment, reweigh the weighing bottle and record the actual mass of BaO used.
6. Plot a graph of temperature against time to obtain a cooling curve. Obtain the maximum temperature change by extrapolation.

Method 1 : Cooling curve

To calculate ΔH_2

Quantity of heat absorbed by solution, $Q = (V_{\text{HNO}_3}) \times 4.18 \times (T_i - T_{\max})$
 $= \text{X J}$

$$\Delta H_2 = -X \div \frac{1.00}{137.0+16.0} / 1000$$

$$= -y \text{ kJ mol}^{-1}$$

Other methods possible: Fixed volume of HNO_3 and change mass of BaO **OR** Fixed mass of BaO and change volume of HNO_3

3 Inorganic and Organic Analysis

(a) 1. Test on FA 7

Test	Observations
(i) To a 1 cm depth of the solution of FA 7 in a test-tube, add a few drops of aqueous barium nitrate followed by dilute nitric acid.	White precipitate formed dissolves in (excess) HNO_3 to form a colourless solution.
(ii) To a 1 cm depth of the solution of FA 7 in a test-tube, add an equal volume of aqueous hydrogen peroxide, then	No effervescence OR a colourless solution is obtained.
add a few drops of aqueous barium nitrate, followed by dilute nitric acid.	White precipitate formed does not dissolve in (excess) HNO_3 .
(iii) To a 2 cm depth of the solution of FA 7 in a test tube, add an equal volume of sulfuric acid, then heat the mixture gently and cautiously.	When heated, gas produced decolourises purple KMnO_4 .
(iv) To a 1 cm depth of the solution of FA 7 in a test-tube, add an equal volume of aqueous sodium hydroxide, then	No ppt formed.
add a few drops of aqueous potassium manganate(VII),	Green solution formed when KMnO_4 added.
followed by sulfuric acid.	Green solution (or yellow or brown) turned colourless on adding sulfuric acid.

- (v) anion: SO_3^{2-}
evidences (any two):

- FA 7 reacts with sulfuric acid / in test (iii) to give SO_2 gas which decolourises KMnO_4 .
- FA 7 formed white precipitate of BaSO_3 formed which dissolves in excess acid.
- The SO_3^{2-} in FA 7 is oxidised to SO_4^{2-} in test (ii) by H_2O_2 which formed white ppt BaSO_4 that is insoluble in excess acid.
- FA7 reduced purple MnO_4^- to green MnO_4^{2-} under alkaline condition OR to colourless Mn^{2+} under acidic condition.

2. (i) Test on FA 8

Test	Observations
(i) To a 1 cm depth of FA 8 in a test-tube, add aqueous sodium hydroxide drop wise till excess. Heat the mixture.	White ppt is soluble in excess NaOH(aq) to form a colourless solution. Gas evolved turned moist red litmus blue.
(ii) To a 1 cm depth of FA 8 in a test-tube, add aqueous ammonia drop wise till excess.	White ppt is soluble in excess $\text{NH}_3(\text{aq})$ to form a colourless solution.

- (ii) Cation: Zn^{2+}
 Evidence: FA 8 forms a white ppt of Zn(OH)_2 which is soluble in excess NaOH to form Zn(OH)_4^{2-} . FA 8 forms a white ppt of Zn(OH)_2 which is soluble in excess NH_3 to form $\text{Zn(NH}_3)_4^{2+}$
 Cation: NH_4^+
 Evidence: FA 8 evolved NH_3 gas upon heating with NaOH(aq) . Hence, NH_4^+ ion is present.

(b) (i) Test on FA 9

Tests	Observation
(i) To a 1 cm depth of FA 9 in a test-tube, add 1 cm^3 of dilute sulfuric acid. Add one drop or 2 drops of aqueous potassium manganate (VII). Place the test-tube in the water bath for several minutes.	Purple KMnO_4 decolourised.
(ii) To a 1 cm depth of FA 9 in a test-tube, add 1 cm^3 (or a few drops) of Fehling's solution.	Brick red ppt formed.

Place the test-tube in the water bath for several minutes.	
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(ii) Functional group in **FA 9** (aliphatic) aldehyde

Evidence: FA 9 undergoes oxidation as it decolorised purple with KMnO_4 . This shows that it cannot be a ketone.

FA 9 undergoes a positive Fehling's test (or oxidation) as it formed a brick red ppt.

Hence, FA9 is an (aliphatic) aldehyde.